

```

# =====
# Function
# =====

check_required_packages <- function(packages) {
  for (pkg in packages) {
    if (!requireNamespace(pkg, quietly = TRUE)) {
      stop(paste0("Package '", pkg, "' is required. Please install it first. "))
    }
  }
}

load_housing_data <- function(file_path) {
  df <- read.csv(file_path, stringsAsFactors = FALSE)

  required_cols <- c("date", "hpi")
  missing_cols <- setdiff(required_cols, names(df))

  if (length(missing_cols) > 0) {
    stop(paste("Missing required columns:", paste(missing_cols, collapse = ",
")))
  }

  df$date <- as.Date(df$date)
  df <- df[order(df$date), ]

  if (!"log_hpi" %in% names(df)) {
    df$log_hpi <- log(df$hpi)
  }

  if (!"diff_log_hpi" %in% names(df)) {
    df$diff_log_hpi <- c(NA, diff(df$log_hpi))
  }

  return(df)
}

make_monthly_ts <- function(values, dates) {
  start_year <- as.integer(format(min(dates), "%Y"))
  start_month <- as.integer(format(min(dates), "%m"))

  ts(
    values,
    start = c(start_year, start_month),
    frequency = 12
  )
}

add_dummy_variables <- function(df) {
  df$post2002 <- ifelse(df$date >= as.Date("2002-01-01"), 1, 0)

  df$crisis <- ifelse(
    df$date >= as.Date("2007-01-01") &
    df$date <= as.Date("2010-12-01"),

```

```

    1,
    0
  )

  return(df)
}

prepare_model_inputs <- function(df) {
  df <- add_dummy_variables(df)

  log_hpi_ts <- make_monthly_ts(
    values = df$log_hpi,
    dates = df$date
  )

  df_diff <- df[!is.na(df$diff_log_hpi), ]

  diff_log_hpi_ts <- make_monthly_ts(
    values = df_diff$diff_log_hpi,
    dates = df_diff$date
  )

  xreg_crisis <- as.matrix(df_diff[, c("crisis"), drop = FALSE])
  xreg_both <- as.matrix(df_diff[, c("post2002", "crisis"), drop = FALSE])

  list(
    df = df,
    df_diff = df_diff,
    log_hpi_ts = log_hpi_ts,
    diff_log_hpi_ts = diff_log_hpi_ts,
    xreg_crisis = xreg_crisis,
    xreg_both = xreg_both
  )
}

fit_candidate_models <- function(inputs) {
  models <- list(
    # Baseline model
    "Random Walk with Drift" = forecast::Arima(
      inputs$log_hpi_ts,
      order = c(0, 1, 0),
      include.drift = TRUE,
      method = "ML"
    ),

    # ARIMA models with drift
    "ARIMA(1,1,0) with drift" = forecast::Arima(
      inputs$log_hpi_ts,
      order = c(1, 1, 0),
      include.drift = TRUE,
      method = "ML"
    ),

    "ARIMA(1,1,1) with drift" = forecast::Arima(
      inputs$log_hpi_ts,

```

```

    order = c(1, 1, 1),
    include.drift = TRUE,
    method = "ML"
  ),

  "ARIMA(2,1,1) with drift" = forecast::Arima(
    inputs$log_hpi_ts,
    order = c(2, 1, 1),
    include.drift = TRUE,
    method = "ML"
  ),

  # SARIMA models with drift
  "SARIMA(1,1,1)(1,0,0)[12] with drift" = forecast::Arima(
    inputs$log_hpi_ts,
    order = c(1, 1, 1),
    seasonal = list(order = c(1, 0, 0), period = 12),
    include.drift = TRUE,
    method = "ML"
  ),

  "SARIMA(1,1,1)(0,0,1)[12] with drift" = forecast::Arima(
    inputs$log_hpi_ts,
    order = c(1, 1, 1),
    seasonal = list(order = c(0, 0, 1), period = 12),
    include.drift = TRUE,
    method = "ML"
  ),

  # ARIMAX models with dummy variables
  # These models use dummy variables to capture changes in average growth.
  "ARIMAX(1,1,1) with crisis dummy" = forecast::Arima(
    inputs$diff_log_hpi_ts,
    order = c(1, 0, 1),
    xreg = inputs$xreg_crisis,
    include.mean = TRUE,
    method = "ML"
  ),

  "ARIMAX(1,1,1) with post-2002 dummy and crisis dummy" = forecast::Arima(
    inputs$diff_log_hpi_ts,
    order = c(1, 0, 1),
    xreg = inputs$xreg_both,
    include.mean = TRUE,
    method = "ML"
  ),

  # SARIMAX models with dummy variables
  # These models add the 12-month dependence structure and use dummy
variables
  # to capture changes in average growth.
  "SARIMAX(1,1,1)(1,0,0)[12] with crisis dummy" = forecast::Arima(
    inputs$diff_log_hpi_ts,
    order = c(1, 0, 1),
    seasonal = list(order = c(1, 0, 0), period = 12),

```

```

    xreg = inputs$xreg_crisis,
    include.mean = TRUE,
    method = "ML"
  ),

  "SARIMAX(1,1,1)(1,0,0)[12] with post-2002 dummy and crisis dummy" =
forecast::Arima(
  inputs$diff_log_hpi_ts,
  order = c(1, 0, 1),
  seasonal = list(order = c(1, 0, 0), period = 12),
  xreg = inputs$xreg_both,
  include.mean = TRUE,
  method = "ML"
)
)

return(models)
}

extract_ic_table <- function(models) {
  table <- do.call(
    rbind,
    lapply(names(models), function(model_name) {
      fit <- models[[model_name]]

      data.frame(
        Model = model_name,
        AIC = as.numeric(AIC(fit)),
        AICc = as.numeric(fit$aicc),
        BIC = as.numeric(BIC(fit)),
        row.names = NULL
      )
    })
  )

  table <- table[order(table$AICc), ]

  return(table)
}

extract_parameter_table <- function(models) {
  table <- do.call(
    rbind,
    lapply(names(models), function(model_name) {
      fit <- models[[model_name]]
      coefs <- stats::coef(fit)

      if (length(coefs) == 0) {
        return(NULL)
      }

      se <- rep(NA_real_, length(coefs))
      names(se) <- names(coefs)

      if (!is.null(fit$var.coef)) {

```

```

    available_se <- sqrt(diag(fit$var.coef))
    se[names(available_se)] <- available_se
  }

  z_value <- coefs / se
  p_value <- 2 * (1 - stats::pnorm(abs(z_value)))

  data.frame(
    Model = model_name,
    Parameter = names(coefs),
    Estimate = as.numeric(coefs),
    Std_Error = as.numeric(se),
    z_value = as.numeric(z_value),
    p_value = as.numeric(p_value),
    row.names = NULL
  )
})
)

return(table)
}

print_model_tables <- function(ic_table, parameter_table) {
  cat("\n===== Information Criteria Table =====\n")
  print(ic_table, row.names = FALSE, digits = 5)

  cat("\n===== Parameter Estimation Table =====\n")
  print(parameter_table, row.names = FALSE, digits = 5)
}

print_key_summaries <- function(models) {
  cat("\n===== ARIMA(1,1,1) with drift Summary =====\n")
  print(summary(models[["ARIMA(1,1,1) with drift"]]))

  cat("\n===== ARIMAX(1,1,1) with post-2002 dummy and crisis dummy Summary  
===== \n")
  print(summary(models[[
    "ARIMAX(1,1,1) with post-2002 dummy and crisis dummy"
  ]]))

  cat("\n===== SARIMAX(1,1,1)(1,0,0)[12] with post-2002 dummy and crisis dummy  
Summary ===== \n")
  print(summary(models[[
    "SARIMAX(1,1,1)(1,0,0)[12] with post-2002 dummy and crisis dummy"
  ]]))
}

run_model_estimation <- function(file_path) {
  set.seed(123)

  check_required_packages(c("forecast"))

  df <- load_housing_data(file_path)
  inputs <- prepare_model_inputs(df)
  models <- fit_candidate_models(inputs)

```

```

ic_table <- extract_ic_table(models)
parameter_table <- extract_parameter_table(models)

print_model_tables(
  ic_table = ic_table,
  parameter_table = parameter_table
)

print_key_summaries(models)

results <- list(
  data = df,
  inputs = inputs,
  models = models,
  ic_table = ic_table,
  parameter_table = parameter_table
)

return(invisible(results))
}

# =====
# Main
# =====

# Run the analysis and print results in the console.
invisible(run_model_estimation(
  file_path = "C:/Users/use/Desktop/NCCU_STAT/Time
Series/final/data/processed/monthly_housing_train.csv"
))

```